

Recipe Site Traffic Prediction

Predicting High-Traffic Recipes for Homepage Selection

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Role: Data Science Project

Dataset: Recipe Site Traffic Dataset

OBJECTIVE

Help Tasty Bytes identify recipes likely to generate high website traffic when featured on the homepage.

- Data-driven homepage selection
- Maximize traffic & subscriptions
- Reduce manual guesswork

Business Problem

Background

Popular recipes increase website traffic by up to 40%

Higher traffic directly drives more subscriptions

Current selection is manual — no systematic approach

Business Goal

Build a predictive model that can:

1

Predict whether a recipe will generate high traffic

2

Support homepage recipe selection decisions

3

Maximize identification of high-performing recipes

Dataset Overview

947

Recipes

8

Original Features

Binary

Target Variable

Features Included

Calories

Carbohydrates

Sugar

Protein

Servings

Category

Recipe ID

High Traffic Indicator

Target Variable

1 = High Traffic

0 = Low Traffic

Data Validation & Cleaning

Data Quality Checks

- ✓ Checked missing values
- ✓ Checked duplicates
- ✓ Validated category values
- ✓ Standardized servings column
- ✓ Created binary target variable

Cleaning Actions

Missing nutritional values filled using median values

Missing traffic values converted to Low Traffic

Servings column converted to numeric format

No duplicate records found

✓ Clean dataset ready for analysis and modeling

Exploratory Data Analysis



574 High Traffic
60.6% of all recipes

373 Low Traffic
39.4% of all recipes

Key Insight

Dataset contains sufficient examples of both classes.
No resampling required for classification modeling.

Relationship Analysis

Category vs Traffic



Traffic Varies

Performance differs significantly across recipe categories.

Top Performers

Certain categories (e.g. Chicken, Breakfast) consistently generate high traffic.

Category Matters

Recipe type is a key predictor — important for model & selection.

Model Development

Problem Type: Binary Classification

Baseline Model

Logistic Regression

Why this model:

Simple & interpretable

Strong linear baseline

Fast to train & validate

Comparison Model

Random Forest

Why this model:

Captures non-linear relationships

Handles feature interactions

Robust ensemble method

Model Evaluation Results

Metric	Logistic Regression	Random Forest
Accuracy	77.37%	73.68%
Precision	84.62%	82.18%
Recall	76.52%	72.17%
F1 Score	80.37%	76.85%
ROC-AUC	86.61%	83.21%

Key Finding

Logistic Regression outperformed Random Forest across all evaluation metrics.

Selected Model



Logistic Regression

Business Impact & Recommendations

Business Metric

High Traffic Capture Rate (Recall)

76.52%

~77 out of every 100 high-traffic recipes correctly identified

Recommendations

- 1 Deploy Logistic Regression as a decision-support tool
- 2 Use model predictions for homepage recipe selection
- 3 Monitor Recall as the primary business metric
- 4 Collect additional user engagement data
- 5 Conduct A/B testing vs. the current manual process

Final Conclusion

The Logistic Regression model provides a reliable, data-driven approach to recipe selection and can help improve homepage traffic and subscription growth for Tasty Bytes.